Redhwan Rahi

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Experience

Software Engineer II **Feb 2024 - Present**

American Express *New York, NY*

- Integrated OpenAI's APIs into an internal search platform used enterprise-wide, adding natural-language querying that **cut average search time by 40%, saving 2,000+ engineering hours annually**
- Led a production database migration from Microsoft SQL Server to **PostgreSQL**, redesigning schema and backend services for full feature parity, **cutting infrastructure costs 50% ($30K+/month)**
- Led adoption of **Claude Code and OpenAI Codex** across internal repositories, building reusable agentic workflows that pull in ticket and codebase context, automate implementation tasks, and support development through testing and review
- Migrated internal search infrastructure from Solr to **Elasticsearch**, re-architecting indexing pipelines for **1M+ records** and building **Kibana** dashboards with vector-based relevance scoring for real-time analytics
- Designed a natural-language-to-SQL AI assistant (React front end, Kotlin backend service) that translates plain-English questions into validated database queries, **eliminating manual data-retrieval requests** for cross-functional stakeholders
- Built an AI code-remediation agent integrated into the developer workflow that auto-applies security patches from a single IDE command, **driving open vulnerabilities to zero** across five production repositories
- Stepped in for the engineering manager during a two-week absence, coordinating **5 developers and 2 QA engineers**, removing delivery blockers, leading engineering touchpoints, and supporting a successful production release

Software Engineer I **Aug 2022 - Feb 2024**

American Express *New York, NY*

- Built a global event scheduling and resource-allocation platform using **Kotlin, Ktor, and SQL**, delivering a reservation workflow that became **72% faster than the legacy process** and supported hundreds of global event reservations
- Spearheaded the migration of a production service to a new internal cloud platform, becoming the **first team in the enterprise to achieve successful blue-green deployments** and establishing a deployment pattern used as a reference for future migrations
- Designed and shipped a **Spring Boot REST API** for automated shipping-label generation that processed **11K+ requests and saved servicing engineers 1,000+ hours** after launch
- **Reduced a production-critical workflow from up to 2 minutes to 200ms** by diagnosing network and backend bottlenecks and restructuring request execution, improving responsiveness for all users
- Built and deployed a **production integration between ServiceNow and an internal microservice platform**, developing a custom UI component and backend service to retrieve and render API data and **establish a new enterprise integration pattern**

Projects

Location-Based Advertising Platform | *TypeScript, Node.js, Fastify, PostgreSQL, Fly.io*

- Founded and built an IoT-driven out-of-home advertising platform delivering hyper-local mobile ad impressions via networked LED devices
- Designed and shipped the full-stack architecture solo: device registration, real-time telemetry ingestion, campaign management, hardware integration, and advertiser-facing analytics dashboards

Technical Skills

Languages: Java, Python, C++, Kotlin, JavaScript/TypeScript, SQL
Backend & Frameworks: Spring Boot, Ktor, Node.js, Fastify, Django, REST APIs
Frontend: React, HTML, CSS
AI & Applied ML: Generative AI, LLM Integration, AI Agents, Agentic Workflows, RAG, Semantic Search, Vector Search, Hugging Face Transformers
Data & Infrastructure: PostgreSQL, Microsoft SQL Server, MongoDB, Redis, Elasticsearch, Kibana, AWS S3, Docker
Engineering: Git, GitHub Actions, Jenkins, CI/CD, System Design, Agile/Scrum

Education

New Jersey Institute of Technology **May 2022**

Bachelor of Science in Computer Science *Newark, NJ*

Honors: Magna Cum Laude